GIT MODULE

Week 9 · Mon

Git & Version Control



Track changes. Collaborate fearlessly. Never lose work.

What is Version Control?



Track Changes

Every save is a snapshot. See exactly what changed, when, and who changed it.



Rollback Anytime

Broke something? Rewind to any previous working state in seconds.



Collaborate Safely

Multiple people edit the same project without overwriting each other.

Git vs GitHub



Git

Free, open-source tool
Runs on YOUR computer
Tracks changes locally
Works offline



GitHub

Cloud hosting platform
Stores your Git repos online
Team collaboration hub
Free for public repos

The Git Workflow



Working Directory

Your project files
(edited but not saved to Git)

```
git add
```



Staging Area

Files queued for
the next commit

```
git commit -m
"..."
```



Local Repository

Permanent history
(all your commits)

Remote Repository (GitHub) ← git push / git pull → teammates

Core Commands
Start & Save

`git init`

Start a new repo

`git status`

See what changed

`git add .`

Stage all changes

`git commit`

Save a snapshot

bash

```
# 1. Create a new local repository
$ git init my-project
$ cd my-project

# 2. Check current state
$ git status
On branch main — nothing to commit

# 3. Stage files
$ git add index.html     # specific file
$ git add .              # all changes

# 4. Commit with message
```

Viewing History

▶ git log

Show all commits

with hash + message

▶ git log --oneline

Compact one-line

per commit

▶ git diff

Show exactly what

lines changed

```
$ git log

commit a3f9b12 (HEAD -> main)

Author: Juma <juma@gachon.ac.kr>

Date: Oct 27 09:15:23 2026

    Add search feature to Book App


$ git log --oneline

a3f9b12 Add search feature to Book App

7c21e44 Fix button color on mobile


$ git diff

- old line (removed)

+ new line (added)
```

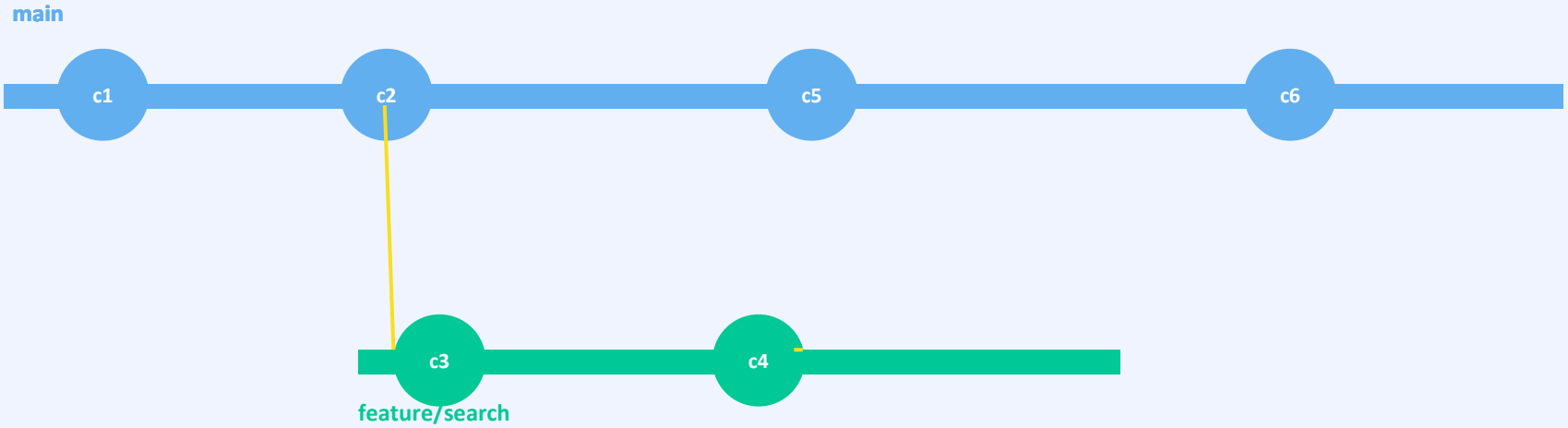
CHAPTER

>

Branching & Merging

Work on features in isolation — merge when ready

Branching — git branch, checkout, merge




```
git checkout -b feature/search
```

Create + switch to new branch

```
git branch
```

List all branches

```
git merge feature/search
```

Merge branch into main

Working with Remote (GitHub)

▶ git remote add

Connect local to GitHub

▶ git push

Upload commits to GitHub

▶ git pull

Download + merge remote changes

▶ git clone

Copy a remote repo locally

```
# Link local repo to GitHub
$ git remote add origin https://github.com/
    username/my-project.git


# Push commits to GitHub
$ git push -u origin main


# Get teammates latest changes
$ git pull origin main


# Copy someone else's repo
$ git clone https://github.com/user/repo
```

GitHub Collaboration Workflow

1 🍴 **Fork**

Copy the repo to your GitHub account

2 💻 **Clone**

Download your fork to your laptop

3 🌿 **Branch**

Create a feature branch

4 🖍️ **Edit & Commit**

Make changes and commit them

5 ☁️ **Push**

Upload your branch to GitHub

6 🔀 **Pull Request**

Ask to merge into the original repo

Live Demo — Follow Along!

Open your terminal now

1
```
mkdir book-app-git && cd book-app-git
```

2
```
git init
```

3
```
echo "Hello Git" > index.html
```

4
```
git status
```

5
```
git add index.html
```

6
```
git commit -m "Initial commit: add index.html"
```

7
```
git log --oneline
```

8
```
git checkout -b feature/search
```

9
```
echo "<input>" >> index.html
```

10
```
git add . && git commit -m "Add search input"
```

11
```
git checkout main
```

12
```
git merge feature/search
```

Summary & Homework

`git init`	Create new repo	`git checkout -b name`	Create + switch branch
`git status`	Check what changed	`git merge branch`	Merge branch into current
`git add .`	Stage all changes	`git push origin main`	Upload to GitHub
`git commit -m "msg"`	Save snapshot	`git pull origin main`	Download from GitHub
`git log --oneline`	View history		

HOMEWORK: Init a local Git repo for your team project. Make 3 commits (add HTML, add CSS, add JS). Push to a new GitHub repo. Share the URL on the Cybercampus by next class.